Pranav Gujjar

AI/ML Engineer | Data Scientist

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PROFESSIONAL SUMMARY

AI/ML Engineer with experience building and deploying production AI systems across **Computer Vision, NLP, Generative AI, Deep Learning, and decision intelligence**. Proven end-to-end delivery across real-time inference, ML pipelines, APIs, analytics, and CI/CD, with solutions achieving **97.34% ROC-AUC, 85.55% NLP accuracy, 80%+ reduction in manual analysis, and 50% faster business decision-making**. MSc Data Science (Distinction), University of East Anglia, UK.

CORE TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, scikit-learn, PyTest), SQL, PySpark, R, MATLAB
ML / DL: Classification, Regression, Clustering, CNNs, RNN, LSTM, GRU, Transformers, TensorFlow, PyTorch
MLOps & Deployment: FastAPI, REST APIs, Docker, GitHub Actions CI/CD, Model Versioning, Artifact Management, Reproducible ML Pipelines, Streamlit
Computer Vision: MediaPipe, Pose Estimation, Biomechanical Analysis, Motion Tracking, OpenCV, U-Net, Dense U-Net, Grad-CAM, Image Segmentation, CLAHE, Patch-Based Learning, IoU, Dice
NLP & Speech: TF-IDF, Linear SVM, Whisper ASR, Rule-Based NLP, Sentiment Analysis, Intent Classification
Generative AI: LLMs, Prompt Engineering, AI Agents, GenAI Workflows, AI-Assisted Coaching, RAG
Data & Analytics: PostgreSQL, MongoDB, Tableau, Power BI, Matplotlib, A/B Testing, Feature Engineering
Cloud / Infra: Supabase, Vercel, Render, Streamlit Cloud, GitHub Actions

EDUCATION

MSc in Data Science — Distinction **Sept 2023 – Sept 2024**
University of East Anglia, UK

Dissertation: Retinal vessel segmentation using U-Net / Dense U-Net on the DRIVE dataset — Sensitivity 79.72%, Accuracy 95.71%, Jaccard Index 67.01%, Dice Coefficient 77.84%, and AUC 97.34%.
Coursework: Deep Learning, Computer Vision, NLP, Time Series, Statistical Modelling.

WORK EXPERIENCE

AI Engineer — ForceField Training.AI - West Midlands,UK **Sept 2026 – Present**

- Engineering and deploying a production **AI movement-intelligence platform** using MediaPipe Pose and real-time Computer Vision for biomechanical assessment, exercise validation, scoring, and personalized coaching.
- Designed **browser-based AI inference** and secure athlete/coach workflows using JavaScript, Supabase, PostgreSQL, RLS, and Vercel, reducing manual movement assessment by **80%+** while avoiding dedicated GPU/model-serving infrastructure.
- Built and optimized AI-driven workout logic across **20+ exercise modules and 31 progressive challenge levels**, integrating real-time pose detection, adaptive feedback, progression tracking, analytics, and production CI/CD.

AI/ML Intern — DR.VIKI (Nurtureheal Healthcare Private Limited), Bengaluru May 2026 – Sept 2026

- Developing production **healthcare AI and Computer Vision systems** for real-time biomechanical analysis, movement assessment, and personalized coaching.
- Engineering end-to-end AI workflows from real-time inference and secure data architecture to analytics and deployment, reducing manual movement evaluation by **80%+**.

Data Science Intern — AI Variant, Bengaluru **Nov 2025 – Feb 2026**

- Delivered two production AI/ML systems spanning **NLP, customer segmentation, REST APIs, analytics dashboards, and cloud deployment**.
- Implemented reproducible ML pipelines and CI/CD using Python, scikit-learn, FastAPI, Streamlit, and GitHub Actions, enabling **50% faster analysis cycles**.

Data Science Intern — Vertexblue Pvt Ltd, India **Jun 2022 – Dec 2022**

- Improved forecasting accuracy by **15%** using ARIMA and RNN-based time-series models in Python; delivered an analytics pipeline contributing to **10%+ cost reduction** and reduced manual processing overhead by **30%** through automated reporting workflows.

PRODUCTION AI PROJECTS

1. ForceField — Real-Time Healthcare AI & Movement Intelligence Platform

Objective:
Built a production-grade healthcare AI platform for real-time biomechanical analysis, movement assessment, and personalized coaching.

Tools Used:
MediaPipe Pose, JavaScript, Computer Vision, Supabase, PostgreSQL, RLS, Vercel, Resend, Git/GitHub

Business Need:
Manual movement assessment was time-intensive and difficult to scale, while server-side AI inference would increase infrastructure cost and latency.

Solution (Applied AI):

- Engineered real-time **pose estimation and biomechanical analysis** across 20+ exercise modules, including rep/hold detection, movement validation, scoring, and adaptive voice coaching.
- Designed **browser-based AI inference**, processing movement data on-device without dedicated GPU/model-serving infrastructure.
- Built secure production workflows for athletes, coaches, and admins using PostgreSQL, Auth/RLS, analytics, progression tracking, and CI/CD.

Impact:

- Reduced manual movement assessment effort by **80%+** through real-time AI automation.
- Supported **20+ exercises and 31 progressive challenge levels** with automated assessment and personalized feedback.
- Kept initial production infrastructure spend at **approximately US$45/month** through client-side inference and serverless/managed services instead of dedicated AI compute infrastructure.

2. Strategic Intelligence Stack | Lead (Team of 5)

Objective:
Built a production-grade customer segmentation and decision intelligence system to convert raw data into actionable business strategies.

Tools Used:
Python, Pandas, NumPy, Scikit-learn (K-Means), FastAPI, Next.js, Vercel, Render, GitHub Actions (CI/CD)

Business Need:
Static segmentation with low usability, no scenario testing, and high manual effort in analysis.

Solution (Data Science):

- Led a team of 5 to build a deterministic **K-Means segmentation pipeline** with reproducible preprocessing and run-based architecture.
- Engineered business insight layers for revenue mix, discount sensitivity, and channel preference, translating customer segments into actionable strategies.
- Built a **real-time what-if simulation engine**, executive dashboards, and GitHub Actions CI/CD with automated deployment to Vercel/Render.

Impact:

- Improved campaign targeting by **20–25%**, reduced analysis time by **30–35%**, and accelerated decision cycles by **40%** (days to hours).
- Increased stakeholder adoption by **1.5–2x** and reduced deployment errors by **25%** through CI/CD automation.

3. ReviewSense AI | Sentiment Intelligence Platform

Objective:
Built a trust-aware NLP system to convert customer reviews into actionable insights and risk signals.

Tools Used:
Python, Scikit-learn (TF-IDF, SVM, Calibration), Pandas, Streamlit, Plotly, Joblib, GitHub Actions (CI/CD)

Business Need:
Unstructured reviews lacked reliable sentiment, risk detection, and prioritization, leading to high manual effort.

Solution:

- Developed a calibrated **TF-IDF + SVM sentiment model**, achieving **85.55% accuracy and 0.853 F1 score**.
- Engineered confidence/risk scoring and detection for negation, mixed sentiment, ambiguity, and other difficult reviews.
- Built an interactive Streamlit analytics dashboard with GitHub Actions CI/CD for automated testing and deployment.

Impact:

- Improved review triaging efficiency by **25%** and reduced manual review effort by **30%** through confidence-based prioritization.
- Increased model adoption by **1.5x** and improved risky-review detection accuracy by **20%**.

DEPLOYED ML SYSTEMS

Retina-AI — Diabetic Retinopathy Screening Platform
PyTorch | Grad-CAM | FastAPI | Streamlit | GitHub Actions

- Built a clinical AI screening system with DR / No-DR classification, Grad-CAM explainability, confidence gating, risk stratification, and automated PDF report generation.
- Developed the full workflow including patient registry, role-based authentication, screening, and report management; deployed with GitHub Actions CI and Ruff linting on Streamlit Cloud.

Identity Vault — Google OAuth Identity Registry
FastAPI | Next.js | PostgreSQL | Google OAuth | Render | Vercel | CI/CD

- Built a deployed identity registry that converts verified Google profiles into searchable internal identity records, reducing manual verification from 2–3 steps to **1 generated-ID lookup**.
- Delivered **7 FastAPI endpoints, 5 stored identity fields**, PostgreSQL persistence, Vercel/Render/Neon deployment, and GitHub Actions CI with **4 backend tests**, Ruff checks, and frontend build validation.

ACADEMIC PROJECTS

Retinal Image Segmentation — MSc Dissertation
TensorFlow | U-Net | Dense U-Net | DRIVE Dataset

- Achieved **97.34% ROC-AUC** for retinal vessel segmentation; built a medical-image preprocessing pipeline using CLAHE, patch-based learning, and augmentation; evaluated using Dice, IoU, Sensitivity, and Specificity.

AI Chatbot — UK Train Ticketing (NLP)
Python | Random Forest | Gradient Boosting | scikit-learn

- Developed intent classification and entity extraction over **400K+ time-series records**; benchmarked Linear, Random Forest, Gradient Boosting, and KNN models with modular reusable preprocessing pipelines and persisted artefacts.